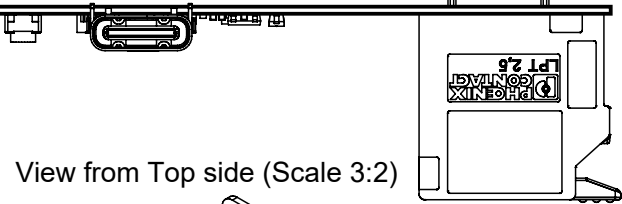
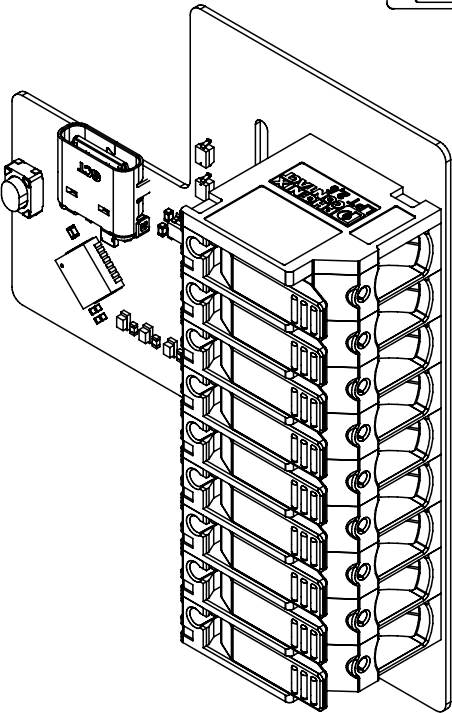


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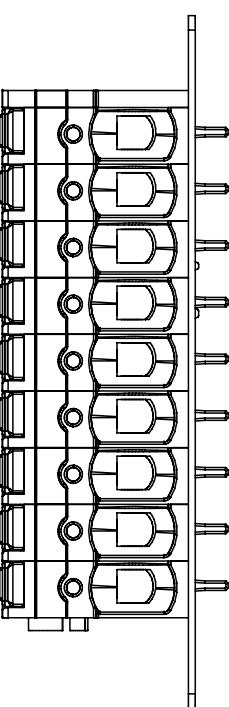
View from Back side (Scale 3:2)



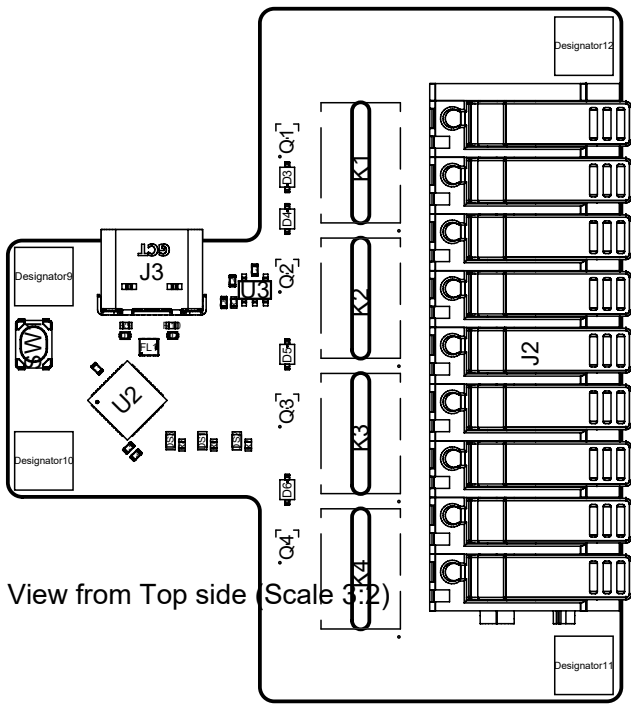
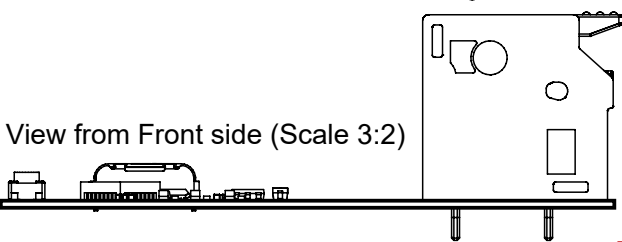
View from Top side (Scale 3:2)



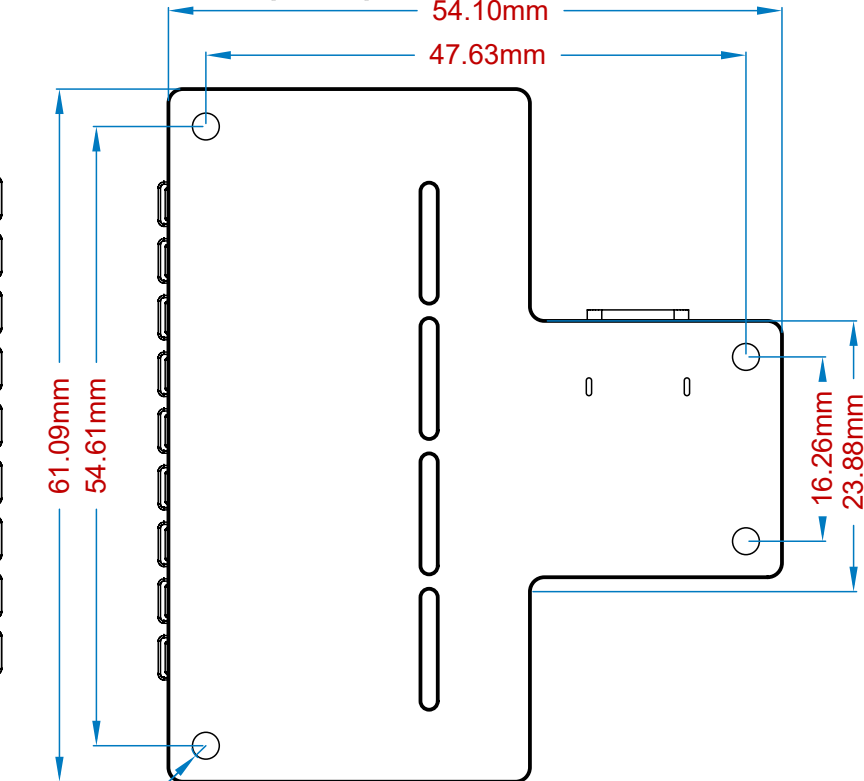
View from Right side (Scale 3:2)



View from Front side (Scale 3:2)



View from Top side (Scale 3:2)



View from Bottom side (Scale 3:2)



College of Engineering –
Center for Environmental
Research and Technology

ASSEMBLY NOTES

Board Name: 242L Signal Controller Trigger Interface

Rev: 1.0

1. Assemble in accordance with IPC-A-610, current revision, Class 2.
2. Solder electrical connections per latest revision of IPC J-STD-001.
3. This assembly contains ESD sensitive components. Handle per ANSI/ESD S20.20.
4. RoHS compliance required: Yes.
5. Mark with current assembly revision.
6. Mount components with polarity and orientation as shown on component designators/silkscreen.

Bill Of Materials

Line #	Designator	Name	Quantity
	C3, C6, C7	GRM155Z71A105KE01D	3
	C4, C5	04025C104KAT4A	2
	C8	GRM155R61A106ME11J	1
	D1, D2	CPDQ5V0	2
	D3, D4, D5, D6	1SS367	4
	DS1, DS2, DS3	LTST-C190GKT	3
	FL1	ECMF02-2AMX6	1
	J1	GRPB052VWVN-RC	1
	J2	1190305	1
	J3	USB4105-GF-A	1
	K1, K2, K3, K4	G6L-1F-DC5	4
	Q1, Q2, Q3, Q4	2SK3018-TP	4
	R3	100KR2F	1
	R6, R8	CRCW040259R0FKED	2
	R7	ERJ-2RK1430X	1
	R9, R10	MCS04020C5101FE000	2
	SW1	TL3365AF180QG	1
	U2	ATSAMD21E18A-MU	1
	U3	NCP718BSN330T1G	1



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APPLICATION		SCALE:		
		FILE NAME: CALTRANS Signal Interface.PCBDwf		
		SHEET: 1 OF 5		

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REV STATUS OF SHEETS	REV											REVISIONS				
	SHEET											ZONE	REV	DESCRIPTION	DATE	APPROVED

Layer Stack Legend

Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.01mm	Solder Resist	Solder Mask	GTS
Copper	Components	0.04mm		Signal	GTL
Prepreg		0.07mm	PP-006	Dielectric	
CF-004	GND	0.04mm		Internal Plane	GP1
		0.32mm	FR-4	Dielectric	
CF-004	PWR	0.04mm		Internal Plane	GP2
Prepreg		0.07mm	PP-006	Dielectric	
Copper	Signals	0.04mm		Signal	GBL
Surface Material	Bottom Solder	0.01mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO

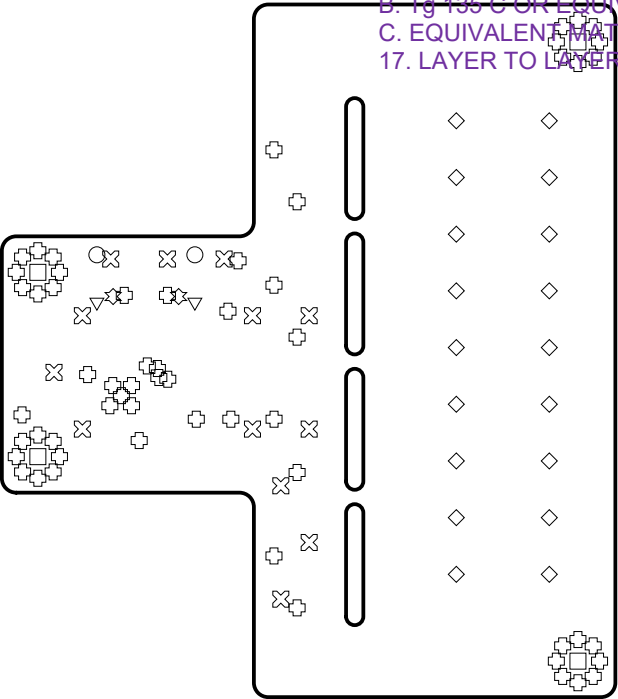
Total thickness: 0.62mm

NOTES: UNLESS OTHERWISE SPECIFIED.

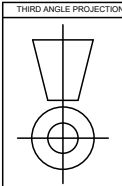
1. FABRICATE PER IPC-6012A CLASS 3.
2. FOR BOARD THICKNESS AND IMPEDANCE DETAILS REFER STACKUP DOCUMENT.
3. PRINTED WIRING BOARD SHALL COMPLY WITH REQUIREMENTS OF ANSI/J-STD-003.
4. SURFACE FINISH: IMMERSION GOLD
5. SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
6. SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
7. THIS PRINTED WIRING BOARD IS DESIGNED WITH A MINIMUM CONDUCTOR WIDTH AND SPACING OF 6 MIL & 4 MILS.
8. ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
9. ALL VIAS ON PAD SHOULD BE FILLED WITH NON CONDUCTIVE EPOXY AND SURFACE SHOULD BE FLAT. FLATNESS TOLERANCE FOR VIA ON PADS: +0.000 /- 0.001 INCHES ON BOTH SIDES. THE MANUFACTURER IS REQUESTED TO SIZE PER THEIR SOLDERMASK TOLERANCE.
10. SOLDER MASK OPENING IS KEPT SAME SIZE AS PAD (1:1) FOR ALL COMPONENTS
11. VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
12. MANUFACTURER'S IDENTIFICATION, DATECODE LETTER SHALL BE SILKSCREENED ON SOLDER SIDE OF THE BOARD.
13. TRACE WIDTH SHOULD BE ACCURATELY ETCHED. MAX TOLERANCE +/- 1 MIL
14. ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED.
15. FLATNESS REQUIREMENTS:
- A. BOW AND TWIST OF ASSEMBLY SUB-PANEL OR SINGULATED PWB SHALL NOT EXCEED 0.7% OF LONGEST SIDE
- B. TEST IN ACCORDANCE WITH THE CURRENT REVISION OF IPC-TM-650 2.4.22
16. PCB MATERIAL REQUIREMENTS:
- A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS. PCB MUST BEAR THE UL94V-0 REGISTERED MATERIAL ID NUMBER.
- B. Tg 135 C OR EQUIVALENT.
- C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CE-CERT.
17. LAYER TO LAYER REGISTRATION SHALL BE WITHIN +/-2 MIL

Drill Table

Symbol	Count	Hole Size	Plated	Hole Tolerance
⊗	13	0.20mm	Plated	
⊕	58	0.30mm	Plated	+/-0.08mm
☆	2	0.65mm	Non-Plated	
◇	18	1.30mm	Plated	
○	2	1.40mm	Plated	
▽	2	1.70mm	Plated	
□	4	2.50mm	Plated	
	99 Total			



Drill Drawing View (Scale 3:2)



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CHECKER: =PCB_CHECKER

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DATE

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PCB_DESIGNER

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Reference Documents

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SIZE: B CAGE CODE: =CAGE_CODE

SCALE: FILE NAME: CALTRANS Signal Interface.PCBDwf

DESIGN ITEM: .Item

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DWG NO: REV:

SHEET: 2 OF 5

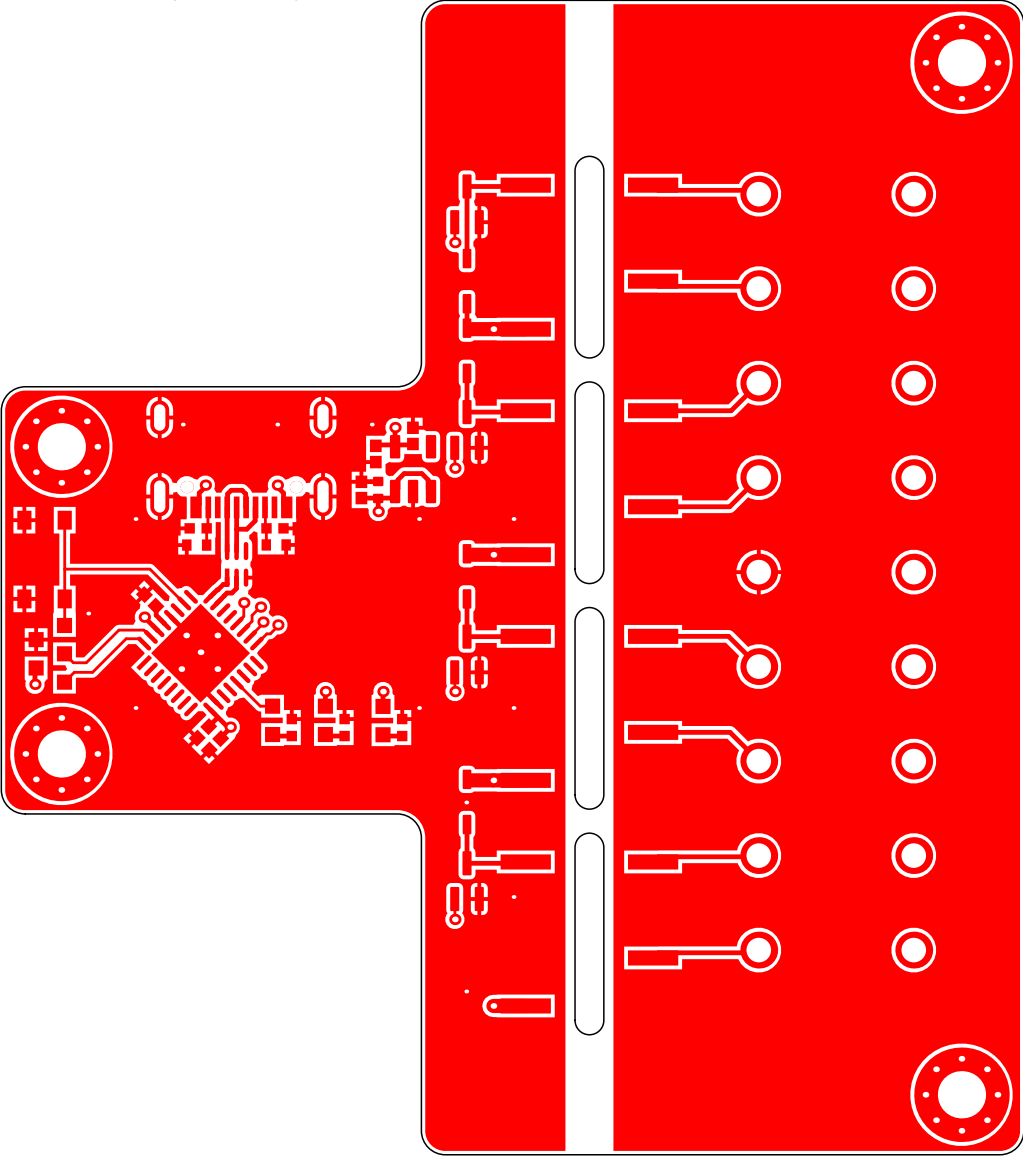
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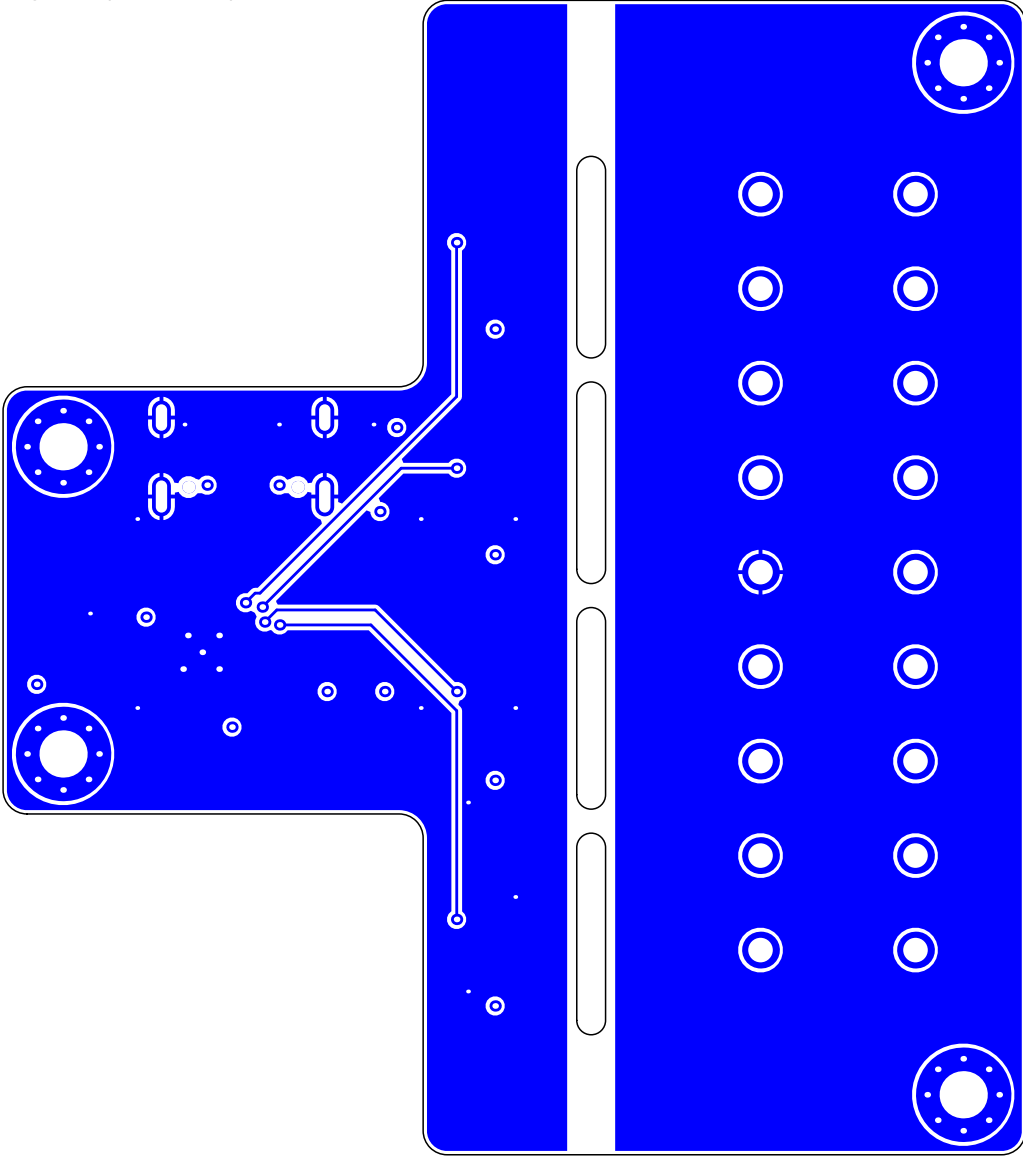
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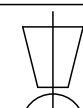
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Components (Scale 5:2)



Signals (Scale 5:2)

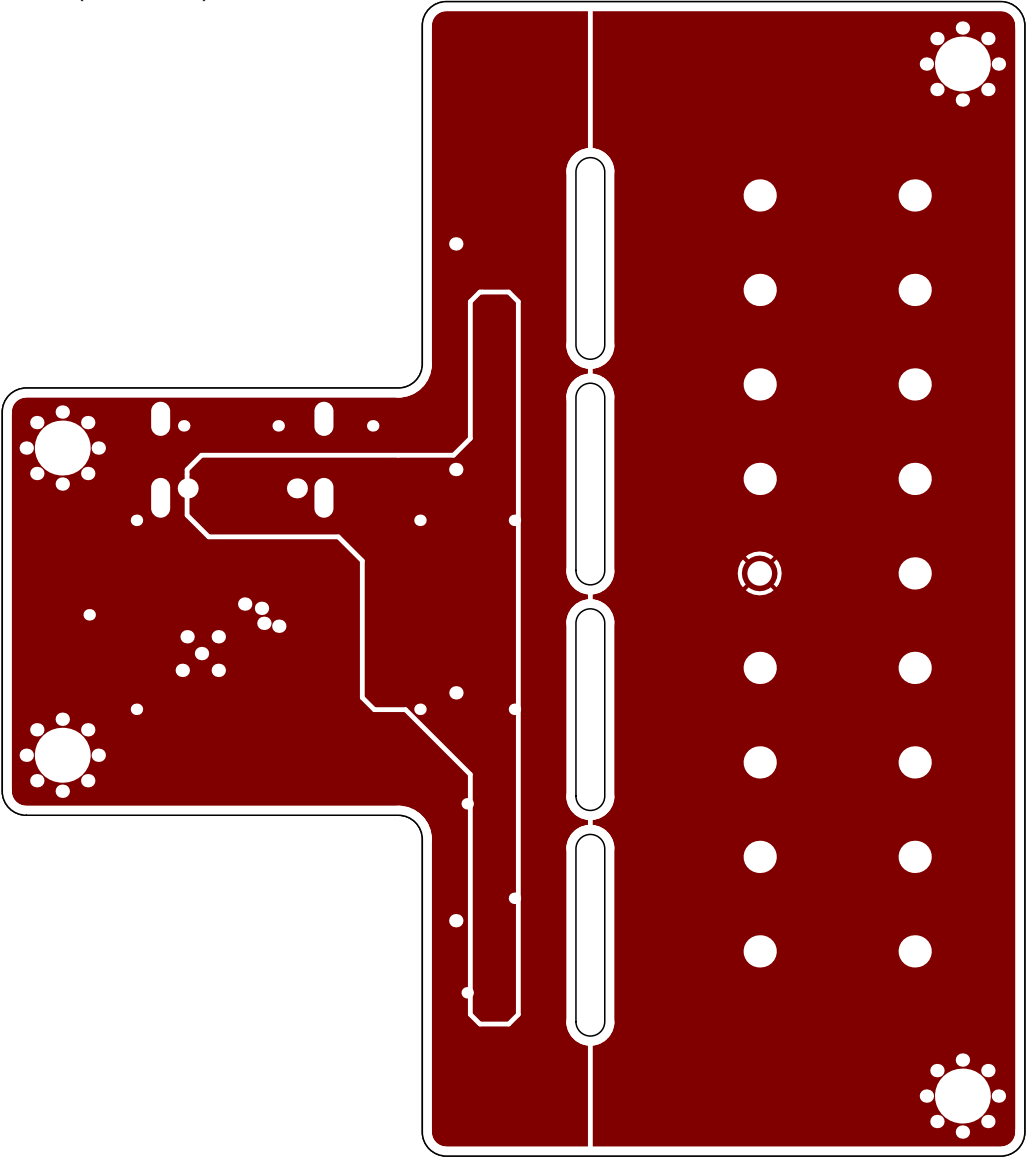


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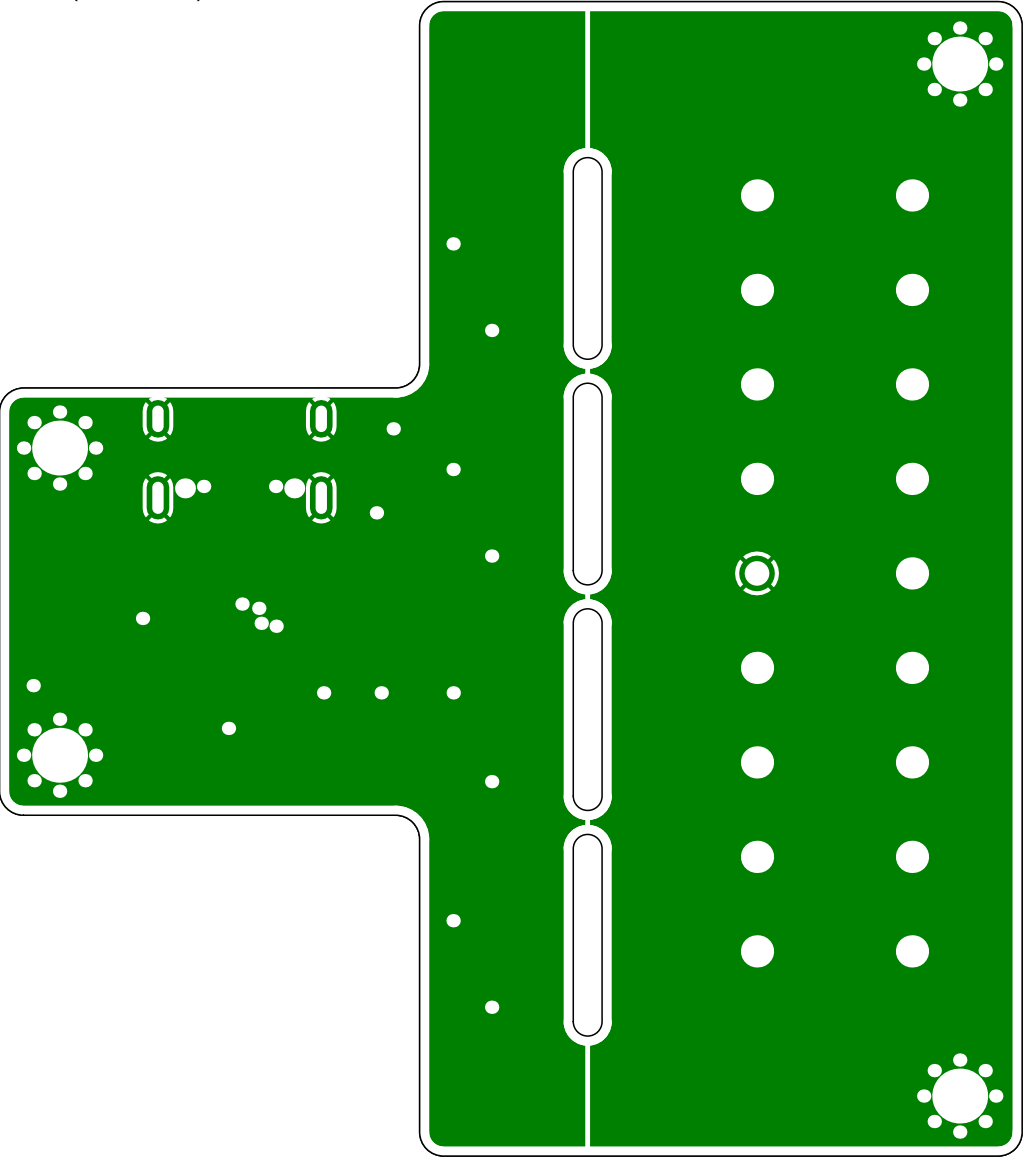
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REV STATUS OF SHEETS	REV											REVISIONS				
	SHEET											ZONE	REV	DESCRIPTION	DATE	APPROVED

PWR (Scale 5:2)



GND (Scale 5:2)

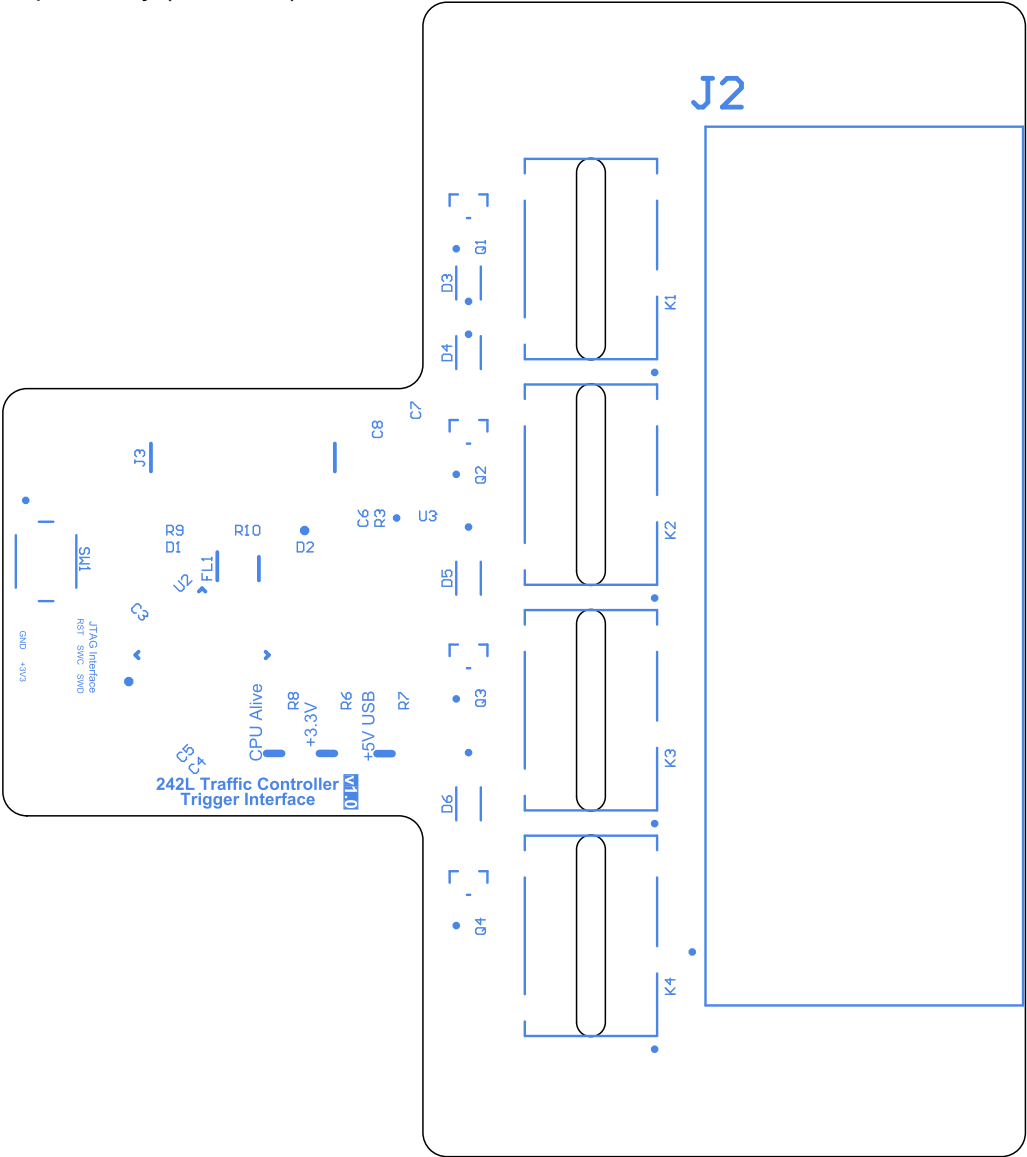


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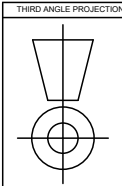
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Top Overlay (Scale 5:2)



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CALTRANS Signal Interface.PCBDwf					



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